

CloudGuard - Stakeholder Validation Checklist

Azure Incident Log Pipeline | Multi-Agent AI System for Banking Infrastructure | Generated: 28-Jun-2026 16:36

A. Network & Infrastructure Monitoring (VP - Networking)							
1	Azure Front Door Log Ingestion	Verify that CloudGuard pulls access and WAF logs from Azure Front Door via Azure Monitor REST API. Confirm logs include client IP, HTTP status code, URL path, routing rule, and time-taken-ms.	VP - Networking				
2	Azure Application Gateway Log Ingestion	Verify that L7 Application Gateway access logs (HTTP method, URL, response code, backend latency) are collected and stored in the CloudGuard database.	VP - Networking				
3	Azure API Management (APIM) Log Ingestion	Confirm APIM gateway logs are ingested with fields such as API operation, response code, backend time, and subscription key identifier.	VP - Networking				
4	Azure VM Diagnostic Log Ingestion	Verify that Azure VM diagnostic / activity logs are collected from the configured VM resource ID.	VP - Networking				
5	Real-Time Service Health Checks	Validate that the Health Check module pings Front Door, App Gateway, APIM, App Service, Blob Storage, WAF, and Log Analytics endpoints in real time and shows HTTP status code, response time (ms), and healthy/unhealthy/unknown status.	VP - Networking				
6	Infrastructure Flow Diagram	Confirm the dashboard displays an interactive infrastructure topology diagram showing the flow: Front Door > App Gateway > APIM > App Service with live health indicators.	VP - Networking				
7	Network Latency & Response Time Tracking	Verify that log preprocessing extracts and normalises time_taken and backend latency fields, and that these are surfaced in analytics charts (e.g., Golden Signals - Latency).	VP - Networking				
B. Cloud Operations & Platform (VP - Cloud)							
8	Azure Monitor REST API Integration	Validate that the Log Collector agent authenticates via Azure AD Service Principal (Tenant ID, Client ID, Client Secret) and pulls activity logs from Azure Monitor for all configured resource IDs.	VP - Cloud				
9	Azure PostgreSQL Database Connectivity	Confirm the application connects to Azure Database for PostgreSQL (async via asyncpg) and that all tables (RawLog, CloudLog, Incident, AgentRun, RootCauseAnalysis, IncidentGroup) are created successfully on startup.	VP - Cloud				
10	Automated Pipeline Scheduler	Verify the APScheduler-based scheduler runs the 9-agent pipeline automatically every N hours (configurable via SCHEDULER_INTERVAL_HOURS). Confirm enable/disable toggle works.	VP - Cloud				

11	Manual Pipeline Trigger	Validate that the Run Pipeline button on the dashboard triggers the full 9-agent pipeline on demand and returns results.	VP - Cloud				
12	Pipeline Enable / Disable Toggle	Confirm the pipeline master switch (PIPELINE_ENABLED) can be toggled from the dashboard UI and the Settings page, and that a disabled pipeline skips scheduled runs.	VP - Cloud				
13	Docker Compose Deployment	Verify the application deploys successfully on an Azure VM using docker compose up -d with three containers: backend (FastAPI), frontend (React/Nginx), and Nginx reverse proxy.	VP - Cloud				
14	Google Cloud Vertex AI (Gemini) Integration	Confirm the Classification, RCA, Resolution, and Notification agents call Gemini 2.5 Flash via Vertex AI using the configured GCP Service Account Key.	VP - Cloud				
15	Pipeline Run History & Audit Trail	Verify that every pipeline run is logged in the AgentRun table with run ID, trigger type, start time, duration, logs processed, incidents created, P1/P2/P3 counts, emails sent, and status.	VP - Cloud				
16	Log Analytics Workspace Integration	Validate that the Azure Log Analytics Workspace ID is used for health checks and that the endpoint responds correctly.	VP - Cloud				

C. Application & AI Pipeline (VP - Application)

17	9-Agent LangGraph Pipeline Execution	Verify the full pipeline executes all 9 agents in order: Log Collector > Preprocessing > Classification > RCA > Priority > Context > Resolution > Orchestrator > Notification.	VP - Application				
18	Log Preprocessing & Deduplication	Validate that the Preprocessing Engine cleans raw logs, normalises timestamps, extracts structured fields (level, category, source, resource ID), and removes duplicates.	VP - Application				
19	AI-Powered Incident Classification (Gemini)	Confirm the Classification Agent uses Gemini AI to detect incident type, intent, severity, and categorisation from preprocessed log data.	VP - Application				
20	Root Cause Analysis (RCA) Agent	Verify the RCA agent identifies probable root causes with confidence scores, collects evidence, matches historical incidents, and identifies affected components.	VP - Application				
21	Priority Assignment (P1 / P2 / P3)	Validate that the Priority Agent assigns correct priority levels: P1 (Critical), P2 (Warning), P3 (Informational). Verify priority justification is recorded.	VP - Application				
22	Historical Context Lookup	Confirm the Context Agent queries PostgreSQL for similar past incidents and surfaces historical resolution patterns.	VP - Application				
23	AI-Powered Resolution Recommendations	Verify the Resolution Agent generates actionable fix recommendations and runbooks using Gemini AI, including step-by-step remediation instructions.	VP - Application				

24	Orchestrator Summary & Routing	Confirm the Orchestrator Agent aggregates outputs from all previous agents, builds a pipeline summary, and routes notifications based on priority.	VP - Application				
25	Incident Detail Panel (UI)	Validate the incident detail panel shows: classification, priority justification, RCA with confidence score, historical context, AI-generated resolution, and runbook.	VP - Application				
26	Real-Time WebSocket Updates	Confirm the dashboard receives live pipeline progress updates via WebSocket: node_active, node_complete, pipeline_start, pipeline_complete, pipeline_error events.	VP - Application				
27	Workflow Visualization (React Flow)	Verify the Workflow View page displays the 9-agent pipeline as an interactive graph with animated node status transitions and real-time output result cards.	VP - Application				
28	Dashboard KPI Cards	Validate top-level KPI cards show: Total Incidents, Open Incidents, P1/P2/P3 counts, Resolved count, Avg Resolution Time, Emails Sent, and pipeline run status.	VP - Application				
29	Analytics & Charting	Verify the Analytics page displays: Incident Trend, Service Breakdown, Error Distribution, Notification History, Top Recurring Issues, MTTR, Golden Signals, Root Cause Distribution, and Resolution Metrics.	VP - Application				
30	Log Explorer with Search & Filters	Confirm the Log Explorer page provides paginated log viewing with filters for log level, source service, and free-text search.	VP - Application				
31	Data Export (JSON / CSV / Excel)	Validate that logs and incidents can be exported in JSON, CSV, and XLSX formats via the Export API and download buttons.	VP - Application				
32	Incident Management (CRUD)	Confirm incidents can be listed with pagination, filtered by priority/status/category/service, viewed in detail, and updated (e.g., mark as Resolved).	VP - Application				

D. Security & Compliance (CISO)

33	JWT Authentication & Protected API	Verify that all API endpoints (except /api/auth/login) require a valid JWT Bearer token. Confirm login returns a signed JWT with 24-hour expiry.	CISO				
34	Login / Logout Flow (UI)	Validate the login page authenticates the user and redirects to the dashboard. Confirm logout clears the JWT token and redirects to login.	CISO				
35	Protected Route Enforcement	Confirm that unauthenticated users cannot access Dashboard, Analytics, Workflow, Log Explorer, or Settings pages.	CISO				
36	WAF Log Monitoring	Verify that WAF logs are ingested and WAF-blocked requests (403 status, rule violations) are classified as security incidents with appropriate priority.	CISO				

37	SSL/TLS Certificate Error Detection	Confirm the RCA agent detects SSL/TLS certificate errors (expired, mismatch, handshake failure) from log patterns and raises them as security-relevant incidents.	CISO				
38	P1 Critical Incident Email Alerting	Validate that P1 (Critical) incidents trigger immediate email notifications via the configured channel (SMTP / Azure Communication Service / Microsoft Graph API).	CISO				
39	Priority-Based Notification Routing	Confirm routing: P1 > Immediate email, P2 > Email digest, P3 > Dashboard only (no email).	CISO				
40	Email Notification Audit (DB Records)	Verify that every email sent is recorded in the database with: email_sent flag, email_sent_at timestamp, and email_recipient address.	CISO				
41	CORS Policy Configuration	Confirm CORS origins are restricted to configured allowed origins only and are not set to wildcard * in production.	CISO				
42	Sensitive Credential Management	Verify that all sensitive credentials are stored in .env file and not hardcoded in source code. Confirm .gitignore excludes .env and key files.	CISO				

E. Cross-Functional / Overall System Validation

43	End-to-End Pipeline Smoke Test	Run the full pipeline end-to-end (manual trigger) and verify all 9 agents execute with expected output.	All Stakeholders				
44	System Health API	Verify the root endpoint (GET /) returns application name, version, and healthy status.	All Stakeholders				
45	Dashboard Summary API	Confirm GET /api/dashboard returns combined summary: raw log stats, processed log stats, incident counts, emails sent, avg resolution time, service breakdown, pipeline status.	All Stakeholders				
46	Multi-Notification Channel Support	Verify the notification system supports three channels: SMTP, Azure Communication Service, and Microsoft Graph API.	All Stakeholders				
47	Responsive & Modern UI Design	Confirm the React dashboard is responsive, includes sidebar navigation, header bar, dark-themed charts, animated counters, sparkline graphs, and smooth transitions.	All Stakeholders				
48	Settings Page - Agent Status & History	Verify the Settings page shows pipeline toggle, agent status, and pipeline run history with trigger type, counts, duration, and status.	All Stakeholders				

Validation Summary

Stakeholder Domain	Total Features	Validated (Pass)	Pending / Fail
VP - Networking	7		
VP - Cloud	9		
VP - Application	16		
CISO	10		
Cross-Functional	6		
TOTAL	48		

Stakeholder Sign-Off

Stakeholder	Name	Signature / Approval	Date
VP - Networking			
VP - Cloud			
VP - Application			
CISO			